

ROBERT JACKSON HOOD

Senior Data Engineer

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Results-driven **Senior Data Engineer** with over **9 years** of experience designing scalable data architectures, building robust **ETL** pipelines, and delivering analytics solutions across **healthcare**, **FinTech**, and **SaaS** domains. Proven expertise in **Python**, **SQL**, **Apache Spark**, **Kafka**, and cloud-native platforms (**AWS**, **GCP**), with a strong foundation in data modeling, governance, and compliance (**HIPAA**, **SOX**). Adept at orchestrating workflows with **Airflow** and **dbt**, deploying infrastructure via **Terraform** and **Docker**, and integrating **machine learning models** into production-grade APIs. Skilled in mentoring engineers, optimizing legacy systems, and driving cross-functional collaboration to support real-time analytics, fraud detection, and executive reporting. Recognized for reducing pipeline failures by 90%, accelerating claims processing by 60%, and leading successful cloud migrations and platform modernization initiatives.

SKILLS

- **Language:** Python, SQL, Scala, R, Shell Scripting, Java
- **Cloud Platforms:** AWS (Redshift, Lambda, S3, Glue, EMR), Azure (Data Factory, Azure ML, Synapse), GCP (BigQuery, Dataflow, Pub/Sub)
- **Data Engineering:** ETL/ELT, Data Pipelines, Apache Spark, Apache Kafka, Hive, Airflow, dbt, Flume, Sqoop, NiFi, Beam
- **DevOps & Infrastructure:** Kubernetes, Docker, Terraform, Jenkins, GitHub Actions, CircleCI, GitLab CI/CD
- **Monitoring & Logging:** CloudWatch, Prometheus, Grafana, Datadog, ELK Stack, Airflow Metrics, Splunk
- **Databases:** PostgreSQL, MongoDB, Oracle, Cassandra, MySQL, SQL Server, DynamoDB, Redis, Kafka Schema Registry
- **ML & AI:** MLOps, DVC, Model Deployment, Pandas, Scikit-Learn, PyTorch, TensorFlow, MLflow, XGBoost
- **Data Modeling:** Dimensional Modeling, Star Schema, Snowflake Schema, Data Vault, Normalization/Denormalization
- **Data Warehousing:** Snowflake, BigQuery, Redshift, Synapse Analytics, Vertica
- **Visualization Tools:** Power BI, Tableau, Looker, Apache Superset, Seaborn, Matplotlib, Excel
- **Testing:** Great Expectations, PyTest, Unit Tests for Data Pipelines, Data Quality Checks
- **Streaming:** Apache Kafka, Apache Flink, Amazon Kinesis, Spark Streaming
- **Familiar with:** Web Scraping, TensorFlow Extended (TFX), MLOps Frameworks, Git, Data Governance, Metadata Management, Data Lineage Tools

PROFESSIONAL EXPERIENCE

Senior Data Engineer | i3 Verticals | Nashville, Tennessee | Jul 2021 - Present

- Developed scalable data pipelines using **Python**, **SQL**, and **Apache Spark**, integrating financial transaction data across multiple platforms.
- Orchestrated **ETL** workflows with **Apache Airflow** and **dbt**, combining structured data from **PostgreSQL** and semi-structured logs from **MongoDB**.
- Integrated **Apache Kafka** for real-time event streaming and landed data into **Snowflake**, bridging ingestion and analytics layers.
- Built **RESTful APIs** using **FastAPI** and **Flask**, exposing curated datasets to internal billing and reconciliation systems.
- Provisioned infrastructure with **Terraform** for **Airflow** environments and **Redshift** clusters; deployed **DAGs** via **CI/CD** pipelines.

- Containerized **Spark** jobs using **Docker** and orchestrated them with **Kubernetes**, optimizing compute for monthly financial close workloads.
- Implemented CI/CD pipelines using **GitHub Actions**, integrating **unit tests** and **data validations** with Great Expectations.
- Modeled financial data using **Star Schema** and **Snowflake Schema**, supporting audit trails and compliance reporting.
- Tuned queries in **Redshift** for legacy reporting and **BigQuery** for modern analytics, applying partitioning and clustering strategies.
- Collaborated with data scientists to deploy fraud detection models using **MLflow**, **DVC**, and **Scikit-Learn**, embedding inference into production APIs.
- Built internal dashboards using **React** and **TypeScript**; deployed **Apache Superset** for ad-hoc analytics and data exploration.
- Integrated **Prometheus**, **Grafana**, and **CloudWatch** for observability across microservices and data workflows.
- Refactored legacy **ETL** scripts written in **Shell** and **Perl** into modular **Python** components with full unit test coverage.
- Defined data contracts and schema registry standards for **Kafka** topics, ensuring compatibility across consumers.
- Enforced data governance policies using **AWS Lake Formation**, IAM roles, and encryption-at-rest for sensitive PII.
- Benchmarked **Spark** (AWS EMR), **Flink** (self-hosted), and **Dataflow** (GCP) for large-scale joins, selecting Spark for ecosystem maturity.
- Participated in **Agile** ceremonies, sprint planning, and roadmap alignment with product and compliance teams.
- Led mentoring sessions for junior engineers on cloud-native architecture, **Git workflows**, and **Pythonic** design principles.
- Integrated Looker for product analytics and **Tableau** for executive reporting, blending real-time and batch data sources.
- Managed **Kafka Schema Registry** for Avro-based ingestion pipelines feeding into **Snowflake** staging layers.
- Implemented cross-cloud streaming ingestion from **Amazon Kinesis** to **BigQuery** via custom connectors for unified analytics.
- Authored architecture decision records (ADRs), onboarding guides, and data flow diagrams to support cross-team scalability.
- Recognized for reducing pipeline failures by 90% and improving platform reliability across two fiscal quarters.

Data Engineer | CVS Health | Woonsocket, Rhode Island | Sep 2018 – May 2021

- Developed backend microservices using **Python**, **Flask**, and **SQLAlchemy**, powering eligibility checks and claims adjudication workflows.
- Orchestrated Spark-based **ETL** pipelines using **Apache Airflow**, processing millions of daily records from **Oracle** and **SQL Server** databases.
- Integrated **Apache Kafka** for asynchronous messaging between pharmacy systems and benefits platforms, improving throughput and fault tolerance.
- Designed RESTful APIs using **Flask** and documented them with **OpenAPI (Swagger)**, ensuring **HIPAA**-compliant data exchange.
- Provisioned infrastructure using **Terraform**, deploying services across **AWS EC2**, **Lambda**, and **RDS** environments.
- Implemented **CI/CD pipelines** with **Jenkins**, supporting legacy workflows via **GitLab CI** during migration phases.
- Containerized legacy services using **Docker**, enabling smoother transitions to cloud-native architecture.
- Tuned **Oracle PL/SQL** procedures and optimized indexing strategies to reduce query latency in high-volume claims databases.
- Modeled healthcare data using **Star Schema** and Dimensional Modeling, supporting actuarial and compliance reporting.
- Built dashboards in **Tableau** for clinical teams and **Power BI** for finance stakeholders, visualizing pharmacy utilization and cost trends.
- Monitored infrastructure and service metrics using **Prometheus** and **Grafana**, with custom exporters for API latency tracking.
- Refactored legacy **ETL** scripts written in **Perl** and **Shell** into modular **Python** jobs with full unit test coverage.
- Enforced role-based access controls and encryption-at-rest using **AWS IAM** and **KMS**, ensuring PHI protection.

- Participated in Agile ceremonies and collaborated with cross-functional teams to align sprint goals and release timelines.
- Led internal workshops on **Python** best practices, **Git workflows**, and **secure API** design for junior developers.
- Supported migration from on-prem systems to **AWS**, coordinating data transfer, schema mapping, and service replication.
- Used **Kafka** for event streaming and batch-loaded data into Redshift for operational reporting across pharmacy systems.
- Authored technical documentation, data flow diagrams, and onboarding guides to support platform scalability.
- Benchmarked **Spark** for ETL workloads and **Presto** for ad-hoc querying, comparing performance against legacy **Hive** jobs.
- Collaborated with compliance teams to implement audit logging and data lineage tracking across ETL workflows.
- Recognized for improving claims processing speed by 60% and reducing system downtime through proactive monitoring and refactoring.

Data Analyst | Cognizant | Teaneck, NJ | October 2015 – August 2018

- Analyzed large datasets using **SQL**, **Excel**, and **Python (Pandas, NumPy)** to generate operational insights and support client reporting needs.
- Built automated data extraction workflows using **SSIS** for structured sources and **Python scripts** for custom logic and post-processing, reducing manual effort by over 60%.
- Developed dashboards in **Tableau** for clinical analytics and **Power BI** for finance teams, visualizing **KPIs** such as claims volume, member engagement, and **SLA adherence**.
- Created stored procedures and optimized queries across SQL Server and Oracle, improving report performance and reducing latency.
- Conducted root cause analysis on data anomalies, implementing validation rules and reconciliation logic across ETL pipelines.
- Supported ETL processes by profiling source data, mapping transformations, and validating load accuracy in staging and warehouse layers.
- Documented data dictionaries, lineage diagrams, and reporting workflows to support audit readiness and cross-team transparency.
- Assisted in migrating legacy Crystal Reports to Tableau, modernizing visualizations and reducing maintenance overhead.
- Delivered ad-hoc analysis using **Excel pivot tables**, **Power Query**, and **VLOOKUP**, enabling faster decision-making for client-facing teams.
- Participated in Agile ceremonies and collaborated with developers and QA teams to ensure data integrity across iterative releases.
- Ensured data handling aligned with **HIPAA** for healthcare clients and **SOX** for financial services clients, supporting regulatory compliance.
- Created KPI scorecards and executive summaries for leadership teams, highlighting trends in operational efficiency and cost performance.
- Recognized for improving report delivery timelines by 40% and reducing manual data prep through automation and process redesign.

EDUCATION

Bachelor of Science | Carnegie Mellon University | Pittsburgh, Pennsylvania | August 2011 – May 2015

Major: Computer Science